Laws of Indices - Mark scheme

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

Write 5^{-2} as a fraction.

A negative power means the reciprocal

Answer: 1/25

Question 2 *Practice question written for this worksheet (not from a past paper)*

Work out $64^{\frac{1}{3}}$

Cube root of 64

Answer: 4

Question 3 *Practice question written for this worksheet (not from a past paper)*

Work out $27^{\frac{2}{3}}$

Cube root: 3; then square: 3^2

Answer: 9

Question 4 *Practice question written for this worksheet (not from a past paper)*

Work out $\left(\frac{1}{4}\right)^{-2}$

Flip the fraction: 4^2

Answer: 16

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Question 5 *Practice question written for this worksheet (not from a past paper)*

Work out $16^{-\frac{1}{2}}$

$\frac{1}{16^{1/2}} = \frac{1}{4}$

Answer: 1/4

Question 6 *Practice question written for this worksheet (not from a past paper)*

Simplify $(27x^6)^{\frac{1}{3}}$

Cube root of 27 is 3; $x^{6 \times \frac{1}{3}} = x^2$

Answer: $3x^2$

Question 7 *Practice question written for this worksheet (not from a past paper)*

Simplify $2x^2y \times 3xy^4$

$2 \times 3 = 6$; x^{2+1}; y^{1+4}

Answer: $6x^3y^5$

Question 8 *Practice question written for this worksheet (not from a past paper)*

Write $\frac{1}{\sqrt{x}}$ as a power of x.

$\sqrt{x} = x^{1/2}$; reciprocal gives a negative power

Answer: x^(-1/2)

Question 9 *Practice question written for this worksheet (not from a past paper)*

Solve $n^4 = 16$

$2^4 = 16$ and $(-2)^4 = 16$

Answer: n = 2 or n = -2

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Laws of Indices - Mark scheme

Question 10 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 6*

Question	Working	Answer	Mark	Notes
6		$64x^{15}y^{18}$	2	B2 for $64x^{15}y^{18}$ (B1 a single product with 2 of 64, x^{15}, y^{18} correct eg $64x^{15}y^{n}$ where $n \neq 18$ **or** $64x^{m}y^{18}$ where $m \neq 15$ **or** $px^{15}y^{18}$ where $p \neq 64$. One term can be missing with 2 correct for B1 eg $64x^{15}$ **or** $64y^{18}$ **or** $(4)x^{15}y^{18}$ **or** $ax^{15}by^{18}$ where a and b are integers as long as not added to any other terms)
				Total 2 marks

Question 11 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 7*

7	eg 3^3 or $(3^{-2})\times 3^{-5}$ or $\frac{3^3}{(3^{10})}$ or $\frac{(3^5)}{3^{12}}$ or $\frac{(3^{-2})}{3^5}$ or $3^{-12}(\times 3^5)$ oe or $-2+5-10$ oe or $-12+5$ oe or $3-10$ oe		2	M1 for a correct application of an index rule as a first step or a correct calculation for n
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	-7		A1 Allow 3^{-7}
				Total 2 marks

Question 12 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 7*

7	(a)		1	1	B1 cao
	(b)		-6	1	B1 Allow 3^{-6}
					Total 2 marks

Question 13 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 8*

8	eg $(8^{-2}\times 8^{9}=)\ 8^{-2+9}$ or 8^{7} or 2^{-6+27} or 2^{21} **or** $(8^{-2}\div 8^{10}=)\ 8^{-2-10}$ or 8^{-12} or $\frac{1}{8^{12}}$ or 2^{-6-30} or 2^{-36} or $\frac{1}{2^{36}}$ **or** $(8^{9}\div 8^{10}=)\ 8^{9-10}$ or 8^{-1} or $\frac{1}{8^{(1)}}$ or 2^{27-30} or 2^{-3} or $\frac{1}{2^{3}}$ **or** $(8^{n}\times 8^{10}=)\ 8^{n+10}$ or 2^{3n+30} **OR** $-2+9=n+10$ oe eg $-6+27=3n+30$ oe **OR** $-2+9-10$ oe eg $\frac{-6+27-30}{3}$ oe		2	M1 for one correct application of an index rule (must be seen in powers of 8 or correct conversion to powers of 2) this could be after an initial mistake – working will need to be clearly seen **OR** for forming a correct equation in the indices alone **OR** for a complete method for the value of n
	Correct answer scores full marks (unless from obvious incorrect working)	-3		A1 Accept 8^{-3} or $(2^{3})^{-3}$
				Total 2 marks

Question 14 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 10*

10	(a)		0	1	B1 condone 150^{0}
	(b)		-2	1	B1 condone 3^{-2}
					Total 2 marks

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Question 15 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 12*

12	(a)		$\frac{9}{10}$	1	B1 oe
	(b)		-6	1	B1
					Total 2 marks

Question 16 *Practice question written for this worksheet (not from a past paper)*

Solve $3^x = \frac{1}{9}$

$\frac{1}{9} = 3^{-2}$

Answer: x = -2

Question 17 *Practice question written for this worksheet (not from a past paper)*

Solve $2^{x+1} = 32$

$32 = 2^5$, so $x + 1 = 5$

Answer: x = 4

Question 18 *Practice question written for this worksheet (not from a past paper)*

Solve $3 \times 4^x = 48$

Divide by 3: $4^x = 16$; $16 = 4^2$

Answer: x = 2

Question 19 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 12*

12	$3^4 = \frac{3^x}{9^{3x}}$ **or** $81 = \frac{3^x}{(3^2)^{3x}}$	$9^2 = \frac{3^x}{9^{3x}}$ **or** $81 = \frac{(9^{0.5})^x}{9^{3x}}$			M1 replacing 81 with 3^4 **or** 9^{3x} with $(3^2)^{3x}$ (or 3^{6x}) **or** replacing 81 with 9^2 **or** 3^x with $(9^{0.5})^x$ (in an equation)
	eg $4 + 6x = x$ or $4 = x - 2(3x)$ oe	eg $2 = 0.5x - 3x$ oe			M1 a correct equation using powers
			-0.8	3	A1 oe, dep on at least M1
					Total 3 marks

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Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 14

14	eg $3^{2n+3-4}\left(=3^3\times3^{1-2n}\right)$ **or** $3^{2n-1}\left(=3^3\times3^{1-2n}\right)$ **or** $\left(3^{2n+3}=\right)3^7\times3^{1-2n}$ **or** $\left(\frac{3^{2n+3}}{3^4}=\right)3^{4-2n}$ **or** $\left(3^{2n+3}=\right)3^3\times3^{5-2n}$ **or** $\frac{3^{2n}}{3^4}=3^{1-2n}$ (division by 3) *This is not an exhaustive list* **or** $2n+3-4$ (= ….) **or** (… =) $3+1-2n$ or $2n-1$ (=…) **or** (... =) $4-2n$ (no other options for this)		3	M1	For one rule of indices used to correctly combine two or more of the **given** expressions (do not need part in brackets) *(must include algebra)* **or** For one of the 4 expressions shown, providing it is clear that they apply to the LHS or to the RHS
	eg $3^{2n+3-4}=3^{3+1-2n}$ **or** $3^{2n+3}=3^{8-2n}$ **or** $3^{2n-1}=3^{4-2n}$ **or** $2n+3-4=3+1-2n$ oe eg $2n-1=4-2n$			M1	A correct single power of 3 on both sides or a correct equation in n without indices (some students may go straight to this and gain M2)
	Working required	$\frac{5}{4}$		A1	oe dep on M1
					Total 3 marks

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2021 · Question 1

to imply a correct method

Q		Working	Answer	Mark	Notes
1	(a)		9	1	B1 allow 3^9
	(b)		21	1	B1 allow 5^{21}
	(c)	$8+2-p=6$ oe eg $8+2=6+p$ or $7^{8+2-p}=7^6$ oe		2	M1 (or embedded eg $8+2=10$, $10-4=6$)
		Correct answer scores full marks (unless from obvious incorrect working)	4		A1 allow 7^4
					Total 4 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 4

4	(a)		12	1	B1 accept x^{12}
	(b)		5	1	B1 accept y^5
	(c)		$125a^{12}r^6$	2	B2 for $125a^{12}r^6$ (B1 for a product in the form ka^pr^q where 2 from k, p or q are correct eg $5a^{12}r^6$ Allow $125a^{12}$ or $125r^6$ or $a^{12}r^6$ so as long as not added to any other terms)
					Total 4 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2021 · Question 4

4	(a)		$81k^8$	2	B2 B1 for 81 or k^8 seen in their final answer.
	(b)		$7m^4n^6$	2	B2 B1 for $7m^4$ or n^6 in a product with no other terms in m or n
					Total 4 marks

Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 6

6	(a)		m^7	1	B1	
	(b)		8	1	B1	Allow k^8
	(c)		$9x^{12}y^{16}$	2	B2	B1 for a product in the form ax^py^q where 2 from a , p or q are correct eg $3x^{12}y^{16}$ (Allow $9x^{12}$ or $9y^{16}$ or $x^{12}y^{16}$ so as long as not added to any other terms)
						Total 4 marks

Question 25

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2022 · Question 6

6	(a)		1	1	B1
	(b)		6	1	B1
	(c)	$206 + m - 214 = -3$ oe or $\frac{7^{-3}\times 7^{214}}{7^{206}}$ or $\frac{7^{211}}{7^{206}}$ oe		2	M1 allow $7^{206+m-214} = 7^{-3}$ oe (must be in the form $7^x = 7^y$ where x and y are correct expressions)
			5		A1 accept 7^5
					Total 4 marks

Question 26

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 9

9	(a)		8	1	B1
	(b)		11	1	B1 accept x^{11}
	(c)		$8k^6m^{12}$	2	B2 for all correct B1 for two correct from 8 or k^6 or m^{12}
					Total 4 marks

Question 27

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 10

10	(a)		1	1	B1
	(b)		6	1	B1 Accept y^6
	(c)		$125a^{12}c^6$	2	B2 for $125a^{12}c^6$ B1 for a product in the form ka^pc^q where 2 from k, p or q are correct eg $5a^{12}c^6$ or $125a^{12}3c^6$ Accept multiplication signs between terms (Allow $125a^{12}$ or $125c^6$ or $a^{12}c^6$ as long as not added to any other terms)
					Total 4 marks

Question 28

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 11

11	(a)	$2^{\frac{1}{2}}\times 2^4$ **or** eg $2 \times (2^4)^2 = (2^x)^2$ or $2^9 = 2^{2x}$		2	M1	for a correct expression in powers of 2 that is equivalent to 2^x eg $2^{\frac{1}{2}}\times 2^4$ **or** for showing $\sqrt{2} = 2^{\frac{1}{2}}$ **and** $16 = 2^4$ **or** for writing the equation in powers of 2 eg $2 \times (2^4)^2 = (2^x)^2$ or $2^9 = 2^{2x}$
		Working required	$\frac{9}{2}$		A1	**or** 4.5 **or** 4½ dependent on M1
	(b)	$\frac{11^{-30}}{11^4}$ **or** $-30 - 4 = n$ **or** $-30 = n + 4$ oe		2	M1	For $(11^{-6})^5$ written as 11^{-30} in the equation or $(11^{-6})^5 = 11^{-30}$ shown in working **or** a correct equation with indices only (no marks for $3.914....\times 10^{-36}$)
		Working required	-34		A1	dep on M1 (as we have asked for working)
						Total 4 marks

Question 29 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 12*

Question	Working	Answer	Mark	Notes
12 (a)	$2^7\times(2^2)^5=(2^2)^x$ oe eg $2^7\times2^{10}=2^{2x}$ or $\left(4^{\frac{1}{2}}\right)^7\times4^5\left[=(4)^x\right]$ oe or LHS written as 2^{17}		2	M1 writing 4^5 and 4^x as powers of 2 or or writing 2^7 as $4^{3.5}$ oe or writing (LHS) $2^7\times4^5$ as 2^{17}
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	8.5		A1 oe $\frac{17}{2}$, $8\frac{1}{2}$, allow $4^{8.5}$ oe
(b)		$25p^4y^{16}$	2	B2 (award B1 for 2 parts correct must be 25 and not 5^2)
				Total 4 marks

Question 30 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 13*

13 (a)			$\frac{25x^4}{y^{10}}$	2	B2 oe for all 3 correct eg $25x^4y^{-10}$ (B1 for a quotient in the form of $\frac{px^q}{y^r}$ eg $\frac{25x^4}{y^6}$ or $\frac{5^2x^4}{y^{10}}$ or $\frac{ax^4}{y^{10}}$ where a ($\neq 25$) is a constant or a product in the form px^qy^{-r} eg $25x^4y^{10}$ or ax^4y^{-10} where a ($\neq 25$) is a constant where 2 from p or q or r are correct Allow $25x^4$ or $\frac{x^4}{y^{10}}$ or x^4y^{-10} or $\frac{25}{y^{10}}$ or $25y^{-10}$ etc so long as not added to any other terms)
	(b)	4^{3p} or $4^n=4^{m-3p}$ or $n+3p=m$ oe or $2^{2n}=\frac{2^{2m}}{2^{6p}}$ or $2^{2n}=2^{2m-6p}$ or $2n=2m-6p$ oe		2	M1
		Correct answer scores full marks (unless from obvious incorrect working)	$m-3p$		A1 oe eg $\frac{2m-6p}{2}$ SC B1 for an answer of $\frac{m}{3p}$
					Total 4 marks

Question 31 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 6*

6	(a)		1	1	B1	cao
	(b)	eg $(5^9\div5^{-3}=)\ 5^6$ **or** $(5^9\times5^{-2}=)\ 5^{11}$ **or** $(5^{-3}\times5^{-2}=)\ 5^{-1}$ **or** $(5^k\div5^{-2}=)\ 5^{k-2}$ **or** $9-3=k-2$ oe **or** $9-3=-2$ **or** $9-3+2$		2	M1	for one correct application of an index rule (must be seen in powers of 5) this could be after an initial mistake – working will need to be clearly seen **or** for forming a correct equation in the indices alone **or** for a complete method for the value of k
		Correct answer scores full marks (unless from obvious incorrect working)	8		A1	condone 5^8
	(c)		$8d^{12}e^{15}$	2	B2	for a correct answer (B1 for answer of the form kd^me^n where at least two of $k=8$, $m=12$ and $n=15$ are correct)
						Total 5 marks

CHALLENGE QUESTIONS

Question 32 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 24*

24	2^3 **and** 2^{4x} or $(2^4)^x$		5	M1	for writing 16^x **and** 8 as a power of 2 (or all as powers of 4,8 or 16)
	$n = x^2 + 4x + 3$ oe or $x^2 + 4x + 3 - n = 0$			A1	for writing n in terms of x correct expression implies first M1
	$(n=)(x+2)^2 - 2^2$ ..... oe or $(x=) -2 \pm \sqrt{n+1}$ $(x=)\dfrac{-4 \pm \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe			M1	for a correct first step in completing the square or using the quadratic formula correctly ft their 3 term quadratic
	$(x=) -2 + \sqrt{n+1}$ oe or $(x=)\dfrac{-4 + \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe			A1	for correctly rearranging to make x the subject (must be positive square root)
	Correct answer scores full marks (unless from obvious incorrect working)	$(x=) -2 + \sqrt{n+1}$ **and** $n > 3$		A1	must be positive square root Accept $(x=)\sqrt{n+1} - 2$ oe and $3 < n$ Accept $(x=)\dfrac{-4 + \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe **and** $n > 3$ or $3 < n$
					Total 5 marks

24 ALT	$4^{\frac{1}{2}n}$, $4^{\frac{1}{2}x^2}$, 4^{2x} **and** $4^{\frac{3}{2}}$	$8^{\frac{1}{3}n}$, $8^{\frac{1}{3}x^2}$ **and** $8^{\frac{4}{3}x}$	$16^{\frac{1}{4}n}$, $16^{\frac{1}{4}x^2}$ **and** $16^{\frac{3}{4}}$	5	M1	for all as powers of 4 or 8 or 16
	$n = x^2 + 4x + 3$ oe or $x^2 + 4x + 3 - n = 0$				A1	for writing n in terms of x correct expression implies first M1
	$(n=)(x+2)^2 - 2^2$ ..... oe or $(x=) -2 \pm \sqrt{n+1}$ $(x=)\dfrac{-4 \pm \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe				M1	for a correct first step in completing the square or using the quadratic formula correctly ft their 3 term quadratic
	$(x=) -2 + \sqrt{n+1}$ oe or $(x=)\dfrac{-4 + \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe				A1	for correctly rearranging to make x the subject (must be positive square root)
	Correct answer scores full marks (unless from obvious incorrect working)	$(x=) -2 + \sqrt{n+1}$ **and** $n > 3$			A1	must be positive square root Accept $(x=)\sqrt{n+1} - 2$ oe and $3 < n$ Accept $(x=)\dfrac{-4 + \sqrt{4^2 - 4\times1\times(3-n)}}{2}$ oe **and** $n > 3$ or $3 < n$
						Total 5 marks

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Question 33

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 11

11		$\frac{m^9k^{15}}{125}$	2	B2 oe for all 3 correct eg $125^{-1}m^9k^{15}$ or $\frac{1}{125}m^9k^{15}$ Accept a =9, b = 15 and c = 125 B1 for a quotient in the form of $\frac{m^pk^q}{r}$ or a product in the form $r^{-1}m^pk^q$ where 2 from p or q or r are correct eg $\frac{m^9k^{15}}{25}$ or $125m^9k^{15}$ Allow m^9k^{15} or $\frac{m^9}{125}$ or $125^{-1}m^9$ or $\frac{k^{15}}{125}$ or $125^{-1}k^{15}$ so long as not added to any other terms Accept two from $a = 9$ or $b = 15$ or $c = 125$ Accept $y125^{-1}m^9k^{15}$ or $\frac{ym^9k^{15}}{125}$ where y is constant
				Total 2 marks

Question 34

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 23

23	**For 2 of** $30=5\times6$ or $30=5\times2\times3$ oe (for numerator) or $\sqrt{180}=6\sqrt{5}$ oe or $\sqrt{180}=2\times3\times\sqrt{5}$ (for denominator) or $25^{2x+7}=\left(5^2\right)^{2x+7}$ or $5^{2(2x+7)}$ oe or $\left(\sqrt{5}\right)^{4x+9}=\left(5^{\frac{1}{2}}\right)^{4x+9}$ or $5^{\frac{1}{2}(4x+9)}$ oe or $\frac{30}{\sqrt{180}}=\sqrt{5}$ oe or 5^{4x+15} as the numerator or 5^{2x+5} as the denominator		3	M1	or 5^{4x+15} as the numerator or 5^{2x+5} as the denominator or $30=5\times6$ or $30=5\times2\times3$ oe or $\sqrt{180}=6\sqrt{5}$ oe or $\sqrt{180}=2\times3\times\sqrt{5}$ or $25^{2x+7}=\left(5^2\right)^{2x+7}$ or $5^{2(2x+7)}$ oe or $\left(\sqrt{5}\right)^{4x+9}=\left(5^{\frac{1}{2}}\right)^{4x+9}$ or $5^{\frac{1}{2}(4x+9)}$ oe or
	$\frac{6\times5\times5^{4x+14}}{6\times5^{0.5}\times5^{2x+4.5}}$ oe eg $\frac{5^{4x+15}}{5^{2x+5}}$ or $\frac{\sqrt{5}^{8x+30}}{\sqrt{5}^{4x+10}}$ oe or $\frac{25^{2x+7.5}}{25^{x+2.5}}$ oe			M1	Correct expression in terms of 6 (or 2 and 3) and 5 with indices Some cancellation could have taken place **or** fully correct with $\sqrt{5}$ and powers **or** 25 and powers
	working required	5^{2x+10}		A1	dep on M2 allow $w = 2x + 10$
					Total 3 marks

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Question 35

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 21

21	$12^2 = 2^4 \times 3^2$ **or** $2 \times 12^2 = 2^5 \times 3^2$ oe **or** $\frac{2\times12^2}{3^2}(=32) = 2^5$		5	M1	
	$18^{4n} = (2\times3^2)^{4n}$ **or** $2^{4n}\times3^{2\times4n}$			M1	
	$3n^2 - 14n - 5\ (=0)$			A1	
	e.g. $(3n+1)(n-5)(=0)$ $n = \frac{14 \pm \sqrt{(-14)^2 - (4\times3\times-5)}}{2\times3}$			M1	for solving their 3 term quadratic equation using any correct method - if factorising, allow brackets which expanded give 2 out of 3 terms correct (if using formula or completing the square allow one sign error and some simplification – allow as far as e.g. $\frac{14\pm\sqrt{196+60}}{6}$ oe)
		$-\frac{1}{3}$, 5		A1	Allow −0.33 or better for $-\frac{1}{3}$
					Total 5 marks

Question 36

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2021 · Question 19

19	(a)	eg $(2^3)^2 \times \sqrt[3]{(2^2)^6}$ **or** $(2^3)^2 \times (4)^{\frac{6}{3}}$ **or** $4^3 \times 4^2$ **or** 2^6 or 2^4 seen **or** $2^6 \times 16$ or 64×4^2 or $8^2 \times 4^2$ or $8^2 \times 16$ or 64×16			3	M1 a correct first stage.
		$2^6 \times (2^{12})^{\frac{1}{3}}$ **or** 1024 **or** 32^2 **or** 4^5 or $2^6 \times 2^4$				M1 dep on 1st M mark.
				2^{10}		A1 dependent on first M1 isw if 2^{10} seen but then 10 given as answer.
	(b)	$(n^{-\frac{4}{5}} =) \frac{1}{16}$ or 0.0625 oe	eg $\left(n^{-\frac{1}{5}}\right)^4 = \left(\frac{1}{2}\right)^4$		4	M1 for sight of $\frac{1}{16}$ oe, even if raised to an incorrect power. **or** for algebraic approach, separating out the 4, or 5 or −1 in the power
		$(n=)\ 16^{\frac{5}{4}}$ or $0.0625^{-\frac{5}{4}}$ oe $(n=)\ 2^5$ or $\sqrt[4]{1048576}$ oe or $\frac{1}{0.0625^{\frac{5}{4}}}$ or $\left(\frac{1}{16}\right)^{-\frac{5}{4}}$	eg $(n=)\ \left(\frac{1}{2}\right)^{-5}$			M2 for a correct expression for n (M1 for one correct algebraic stage eg $n^{-\frac{1}{5}} = \frac{1}{2}$)
				32		A1
						Total 7 marks

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